Informatica® Metadata Command Center
October 2026

Amazon S3 Sources

Informatica Metadata Command Center Amazon S3 Sources
October 2026

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Table of Contents

Preface. 5

Chapter 1: Introduction to Amazon S3 catalog sources. 6

Extraction and view process. 7

About the Amazon S3 catalog source. 8

Extracted metadata. 8

Data profiling for Amazon S3 objects. 9

Compatible connectors. 10

Chapter 2: Before you begin. 11

Verify permissions. 11

Permissions to extract metadata. 11

Permissions to run data profiles. 11

Permissions to run data classification. 12

Permissions to run relationship discovery. 12

Permissions to run glossary association. 12

Create a connection. 12

Basic authentication. 14

AssumeRole via EC2 role authentication. 15

Credential Profile File authentication 16

IAM Roles Anywhere authentication. 18

Configure runtime environment for document classification. 20

Enable document classification. 21

Import a relationship inference model. 21

Chapter 3: Create catalog sources in Metadata Command Center. 23

Step 1. Register a catalog source. 23

Step 2. Configure capabilities. 25

Configure metadata extraction. 25

Filter guidelines and examples. 29

Configure data profiling and quality. 31

Configure data observability. 35

Configure data classification. 37

Configure relationship discovery. 38

Configure glossary association. 39

Configure automation. 39

Step 3. Associate stakeholders and asset groups. 40

Step 4. Run or schedule the job. 42

Chapter 4: View results in Data Governance and Catalog. 44

View metadata extraction results. 44
View data lineage. 46
View lineage at the catalog source level 46
View lineage at the data set level. 46
View lineage at the data element level. 47
View data profiling results 47
View data observability results 48
View classified data. 49
View glossary associations. 49

Appendix A: Managing catalog sources. 51

Cloning a catalog source configuration. 51
Deleting or purging a catalog source. 52
Purging obsolete objects from a catalog source. 52
Curating CLAIRE recommendations. 53
Managing configuration permissions for catalog sources. 53

Preface

Read *Amazon S3 Sources* to learn how to register and configure Amazon S3 sources in Metadata Command Center as catalog sources. After you configure a catalog source, you extract metadata and then view the results in Data Governance and Catalog.

CHAPTER 1

Introduction to Amazon S3 catalog sources

You can use Metadata Command Center to extract metadata from a source system.

A source system is any system that contains data or metadata. For example, Amazon S3 is a source system from which you can extract metadata through an Amazon S3 catalog source. A catalog source is an object that represents and contains metadata from the source system.

Before you extract metadata from a source system, you first create and register a catalog source that represents the source system. Then you configure capabilities for the catalog source. A capability is a task that Metadata Command Center can perform, such as metadata extraction, lineage discovery, autonomous enrichment, data profiling, data classification, or glossary association.

When Metadata Command Center extracts metadata, Data Governance and Catalog displays the extracted metadata and its attributes as technical assets. You can then perform tasks such as analyzing the assets, viewing lineage, and creating links between those assets and their business context.

The following table describes the capabilities of the catalog source:

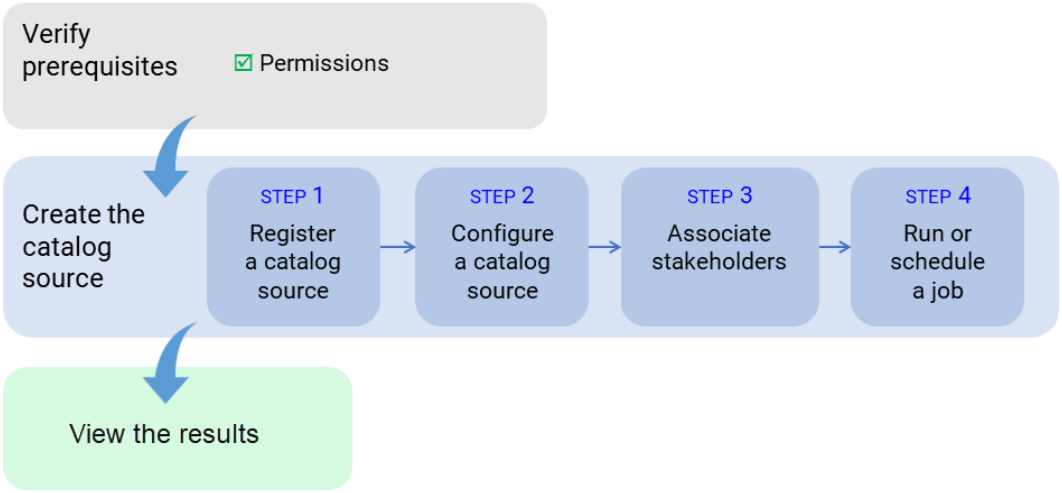
Capability	Description
Incremental metadata extraction	An incremental metadata extraction extracts only the changed and new objects since the last catalog source job run. Incremental metadata extraction doesn't remove deleted objects from the catalog and doesn't extract metadata of code-based objects if applicable.
Serverless Runtime Environment	A serverless runtime environment is an advanced serverless deployment solution that doesn't require downloading, installing, configuring, or maintaining a Secure Agent or Secure Agent group. You can use a serverless runtime environment in the same way that you use a Secure Agent when you configure a catalog source.
Data Profiling and Quality	- **Data Profiling**. Assesses source metadata and analyzes the collected statistics to discover content and structure, such as value distribution, patterns, and data types. - **Data Quality**. Measures the reliability of the data and enables data usage.
Data Observability	Identifies anomalies in the characteristics of the data that you profile.
Data Classification	Data classification is the process of identifying and organizing data into relevant categories based on the functional meaning of the data. Classifying data can help your organization manage risks, compliance, and data security.
Relationship Discovery	The relationship discovery capability identifies pairs of similar columns and relationships between tables within a catalog source.

Capability	Description
Glossary Association	You can associate terms that are in the glossary with technical assets to provide user-friendly business names to technical assets. Glossary Association automatically associates glossary terms with technical assets or recommends glossary terms that you can manually associate with technical assets in Data Governance and Catalog.
Autonomous Enrichment	Automatically propagates classification, description, business name, and glossary enrichments from an asset to related assets based on the lineage between assets. Autonomous enrichment reduces manual curation effort by extending the enrichments that you apply to one asset across assets that are connected to it through lineage.

Extraction and view process

To extract metadata from a source system, configure the catalog source and run the extraction job in Metadata Command Center. Then view the results in Data Governance and Catalog.

The following image shows the process to extract metadata from a source system:



After you verify prerequisites, perform the following tasks to extract metadata from Amazon S3:

1. Register a catalog source. Create a catalog source object, select Amazon S3, and then select and test the connection.
2. Configure the catalog source. Specify the runtime environment and configure parameters for metadata extraction. Optionally, add filters to include or exclude source system assets from metadata extraction. You can also configure other capabilities such as lineage discovery, autonomous enrichment, data profiling and quality, data classification, or glossary association.
3. Optionally, associate stakeholders. Associate users with technical assets, giving the users permission to perform actions determined by their roles.
4. Run or schedule the catalog source job.

After you run the catalog source job, you view the results in Data Governance and Catalog.

About the Amazon S3 catalog source

You can use the Amazon S3 catalog source to extract metadata from an Amazon S3 source system. You can also extract metadata from a third-party storage provider such as Scality RING or MinIO.

Amazon Simple Storage Service (Amazon S3) is an object storage service offered by Amazon Web Services (AWS).

Extracted metadata

You can extract specific metadata from an Amazon S3 source system with an Amazon S3 catalog source.

Metadata Command Center extracts the following metadata from an Amazon S3 source system:

- File
- Folder
- Bucket
- Iceberg Table
- Iceberg Field

You can extract metadata from S3 data sources. By default, you extract metadata from the Amazon S3 Storage, but you can also extract metadata from the S3 Scality Storage or MinIO.

You can extract metadata from the following file types:

- AVRO
- Compressed files
  - TAR
  - ZIP
- Delimited files
  - CSV
  - TSV
- E-mail files
  - EML
  - EMLX
  - MSG
- JSON
- JSONL
- Markdown
- Microsoft Excel files
  - Excel 97-2003 Workbook with XLS extension
  - Excel Workbook with XLSX extension
  - Excel Macro-Enabled Workbook with XLSM extension
- Microsoft Word files
  - DOC
  - DOCX

- ORC
- Presentation files
  - PPT
  - PPTX
- Parquet
- PDF
- Rich Text Format (RTF)
- TXT
- XML
- XSD

Note: Catalog source jobs can extract metadata from protected unstructured files. However, some metadata attributes of such files might not get extracted or displayed in Data Governance and Catalog.

The following table lists the structures associated with the file types that you can extract metadata from:

File Type	Partition structure
AVRO	Single partition, multiple partitions, schema merge
CSV	Single partition, multiple partitions, schema merge
JSON	Single partition, multiple partitions, schema merge
JSONL	Single partition, multiple partitions, schema merge
Parquet	Single partition, multiple partitions, schema merge
XML	Single partition, multiple partitions, schema merge

You can extract workbooks, worksheets, and columns from Microsoft Excel files.

You can extract metadata from XML and XSD file formats as XML file objects. Metadata Command Center extracts only elements and attributes from XML and XSD files. If the size of the XML file exceeds 100 KB, Metadata Command Center extracts metadata from the initial 100 KB of the file. However, for XSD file types, Metadata Command Center extracts complete metadata.

Data profiling for Amazon S3 objects

Configure data profiling to run profiles on the metadata extracted from an Amazon S3 source system. You can view the profiling statistics in Data Governance and Catalog.

You can run data profiles on the following objects:

- Avro
- CSV
- JSON
- Parquet

To run profiles on Avro, JSON, or Parquet files, connect to an advanced cluster.

For more information about advanced clusters, see *Advanced Clusters* in the Cloud Common Services help.

You can also run profiles on a Secure Agent installed on an Amazon EC2 instance on AWS that uses the Instance Metadata Service Version 2 (IMDSv2) method.

The data profiling task runs profiles on the following data types for Avro, CSV, JSON, and Parquet file formats:

File format	Data type
Avro	- INT - STRING
CSV	- STRING
JSON	- ARRAY - DOUBLE - INTEGER - OBJECT - STRING
Parquet	- BOOLEAN - INT32 - INT64 - FLOAT - DOUBLE - DATE - DECIMAL - STRING

Sampling type

You can run the data profiling task on all rows for an Amazon S3 source system.

Compatible connectors

Before you configure an Amazon S3 catalog source, you must connect to the Amazon S3 source system.

Use the Amazon S3 v2 connector to connect to the Amazon S3 source system.

For information about configuring a connection, see *Connections* in the Administrator service.

CHAPTER 2

Before you begin

Before you create a catalog source, ensure that you have the information required to connect to the source system.

Perform the following tasks:

- Assign the required permissions.
- Configure a connection to the Amazon S3 source system.
- Optionally, if you want to identify pairs of similar columns and relationships between tables within a catalog source, import a relationship inference model.

Verify permissions

To extract metadata and to configure other capabilities that a catalog source might include, you need account access and permissions on the source system. The permissions required might vary depending on the capability.

Permissions to extract metadata

Ensure that you have the required permissions to enable metadata extraction.

Grant the following permissions:

- Read permission for the user account that you use to access the catalog source.
- Access permission. Required if the user account is different from the user account that you used to create the Amazon S3 catalog source.

Permissions to run data profiles

Ensure that you have the required permissions to run profiles.

Grant the following permissions:

- ListBucket. Required to view objects from Amazon S3 buckets.
- ListBucketMultipartUploads. Required to list multipart object uploads to Amazon S3 buckets that are in progress.
- GetObject. Required to read objects from Amazon S3 buckets.
- PutObject. Required to process staging data for Avro and Parquet files.
- DeleteObject. Required to delete staging data of Avro and Parquet files.

Permissions to run data classification

You don't need any additional permissions to run data classification.

Permissions to run relationship discovery

You don't need any additional permissions to run relationship discovery.

Permissions to run glossary association

You don't need any additional permissions to run glossary association.

Create a connection

Before you configure the Amazon S3 catalog source, create a connection object in IDMC Administrator.

To use an external secrets manager to connect to the source system, use one of the following authentication tools:

- AWS Secrets Manager
- Azure Key Vault

1. In Administrator, select **Connections**.
2. Click **Add Connection**.
3. Search for and select **Amazon S3 V2** and then click **Next**.
4. Enter the following connection details:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.

Property	Description
Runtime Environment	The name of the runtime environment where you want to run tasks. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Access Key	The access key ID used to access the Amazon account resources. Required if you do not use AWS Identity and Access Management (IAM) authentication. **Note:** Ensure that you have valid AWS credentials before you create a connection.
Secret Key	The secret access key used to access the Amazon account resources. This value is associated with the access key and uniquely identifies the account. You must specify this value if you specify the access key ID. Required if you do not use AWS Identity and Access Management (IAM) authentication.
Folder Path	Bucket name or complete folder path to the Amazon S3 objects. For tasks other than application ingestion and database ingestion tasks, don't use a slash at the end of the folder path. For example, `<bucket name>/<my folder name>`.
IAM Role ARN (Advanced Setting)	The Amazon Resource Name of the IAM role assumed by the IAM user or the EC2 role to use temporary security credentials and access the AWS resources.
External ID (Advanced Setting)	The AssumeRole request parameter to generate temporary session credentials that enable an IAM user or AWS service to assume a role. For more information about how to use the assume role, see the Using an assume role for Amazon S3 resources in Informatica Cloud Data Integration How-To Library.
Use EC2 Role to Assume Role (Advanced Setting)	Enables the EC2 role to assume an IAM role specified in the IAM Role ARN option. **Note:** If you choose to configure the IAM user to assume an IAM role specified in the IAM Role ARN option, clear this option.
S3 Account Type (Advanced Setting)	The type of the Amazon S3 account. Select one of the following options: - Amazon S3 Storage. Enables you to use the Amazon S3 services. - S3 Compatible Storage. Enables you to use the endpoint for a third-party storage provider such as Scality RING or MinIO. Default is Amazon S3 storage.
REST Endpoint (Advanced Setting)	The S3 storage endpoint required for an S3 compatible storage. Enter the S3 storage endpoint in the HTTP or HTTPs format. For example, `http://s3.isv.scality.com`.

5. Select the authentication type to connect to Amazon S3 and enter the required properties.

   You can use the following authentication types:

   - Basic Authentication
   - AssumeRole via EC2 role Authentication
   - Credential Profile File Authentication
   - IAM Roles Anywhere Authentication
6. Click **Test** to test your connection to the source system.
7. Click **Save**.

Basic authentication

Basic authentication requires access key and secret key values from your AWS account.

The following table describes the basic connection properties for basic authentication:

Property	Description
Access Key	Access key to access the Amazon S3 bucket.
Secret Key	Secret key to access the Amazon S3 bucket. The secret key is associated with the access key and uniquely identifies the account.
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>`
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).

AssumeRole via EC2 role authentication

AssumeRole via EC2 role authentication requires you to enable the EC2 role to assume another IAM role specified in the IAM Role ARN option.

The following table describes the basic connection properties for AssumeRole via EC2 role authentication:

Property	Description
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>`
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).

Property	Description
IAM Role ARN	The Amazon Resource Name (ARN) of the AWS Identity and Access Management (IAM) role assumed by the user to use the dynamically generated temporary security credentials. Enter the ARN value if you want to use the temporary security credentials to access AWS resources. **Note:** Even if you remove the IAM role that grants the agent access to the Amazon S3 bucket, the test connection is successful. For more information about how to get the ARN of the IAM role, see the AWS documentation.
External ID	The external ID of your AWS account. External ID provides a more secure access to the Amazon S3 bucket when the Amazon S3 bucket is in a different AWS account.
Use EC2 Role to Assume Role	Enables the EC2 role to assume another IAM role specified in the IAM Role ARN option. By default, this property is not selected. **Note:** The EC2 role must have a policy attached with permissions to assume an IAM role from the same or different account.

Credential Profile File authentication

Credential profile file authentication requires the credential profile file path and profile name.

The following table describes the basic connection properties for credential profile file authentication:

Property	Description
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>`
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).
Other Authentication Type	Determines whether you want to use the credential profile file authentication to connect to Amazon S3. Select one the following authentication types: - NONE. Select if you do not want to use credential profile file authentication. - Credential Profile File Authentication. Select to use credential profile file authentication to access the Amazon S3 credentials from a credential file. **Note:** To use IAM Roles Anywhere authentication also, you must select Credential Profile File Authentication as the authentication type. Enter the credential profile file path and profile name to connect to Amazon S3. You can use permanent IAM credentials or temporary session tokens when you configure the credential profile file authentication. Default is NONE.

Property	Description
Credential Profile File Path	The path to the credential profile file. If you don't enter the credential profile path, the Secure Agent uses the credential profile file available in the following default location in your home directory: `~/.aws/credentials`
Profile Name	Name of the profile in the credential profile file used to get credentials to access Amazon S3 resources. If you don't enter the profile name, the credentials from the default profile in the credential profile file are used.

IAM Roles Anywhere authentication

You can configure AWS IAM Roles Anywhere authentication to generate temporary security credentials to securely access Amazon S3 resources from your servers, applications, or containers outside of AWS using X.509 certificates instead of long-term credentials.

IAM Roles Anywhere authentication requires the credential profile file path and profile name.

The following table describes the basic connection properties for IAM Roles Anywhere authentication:

Property	Description
Folder Path	Amazon S3 bucket name or the folder path within the Amazon S3 bucket where the Amazon S3 objects are stored. For example, `<bucket name>/<my folder name>`
Region Name	The AWS region of the bucket that you want to access. Before you select a region, ensure that the Secure Agent is up and running. If you still do not see your region in the list, restart the Data Integration Server. Select one of the following regions: - Africa (Cape Town) - Asia Pacific (Hong Kong) - Asia Pacific (Hyderabad) - Asia Pacific (Jakarta) - Asia Pacific (Melbourne) - Asia Pacific (Osaka) - Asia Pacific(Mumbai) - Asia Pacific(Seoul) - Asia Pacific(Singapore) - Asia Pacific(Sydney) - Asia Pacific(Tokyo) - AWS GovCloud (US) - AWS GovCloud (US-East) - Canada(Central) - EU (Milan) - EU (Paris) - EU (Spain) - EU (Stockholm) - EU (Zurich) - EU(Frankfurt) - EU(Ireland) - EU(London) - Israel (Tel Aviv) - Middle East (Bahrain) - Middle East (UAE) - South America(Sao Paulo) - US East(N. Virginia) - US East(Ohio) - US ISO East - US ISO West - US ISOB East (Ohio) - US West(N. California) - US West(Oregon) Default is US East(N. Virginia).
Other Authentication Type	Determines whether you want to use the IAM Roles Anywhere authentication to connect to Amazon S3. To use IAM Roles Anywhere authentication, select **Credential Profile File Authentication**. Enter the credential profile file path and profile name to connect to Amazon S3.

Property	Description
Credential Profile File Path	The path to the credential profile file. If you don't enter the credential profile path, the Secure Agent uses the credential profile file available in the following default location in your home directory: `~/.aws/credentials`
Profile Name	Name of the profile in the credential profile file used to get credentials to access Amazon S3 resources. If you don't enter the profile name, the Secure Agent uses the credentials from the default profile in the credential profile file.

Configure runtime environment for document classification

To enable the document classification capability for the catalog source, you must configure an elastic runtime environment and configure at least one GPU node on the runtime environment.

A GPU node is a prerequisite to train the classification engine and run the document classification capability. For better performance during classification training and inference, you can install more than one GPU node.

1. Configure an elastic runtime environment in Administrator.

   For information about how to configure and use the elastic runtime environment, see *Elastic Runtime Environments* in the Administrator help.

2. Download the `gpu_setup.zip` file.

   The ZIP file is available in the following location:

   ```
   <Informatica Secure Agent installation directory>/downloads/package-
   MetadataPlatformService.<version>/package/scripts/gpu_setup.zip
   ```

3. Extract the ZIP file on the master node of your elastic runtime environment installation.

   Run the following command:

   ```
   unzip gpu_setup.zip -d gpu_setup && cd gpu_setup
   ```

4. From the extracted folder, open the `gpu_node.txt` file and configure the following key-value pairs:

Key	Value
JOB_NODE_ASG_NAME	Name of the existing job-node Auto Scaling Group for your elastic runtime environment cluster.
SSH_KEY_PATH	The absolute path to the PEM key that was generated during the elastic runtime cluster installation.
SSH_USER	The SSH user name for connecting to the cluster nodes. Default is cloud-user.
GPU_WORKER_INSTANCE_TYPE	The EC2 instance type for GPU nodes. Default is g5.xlarge. The minimum requirement is g4dn.xlarge.

5. Run the preflight check.

Use the following command:

```
./gpu-setup-preflight.sh
```

Ensure that there are no failures in the summary. Review the warnings.

6. Run the installer to install the GPU node on the runtime environment.

   Use the following command:

   ```
   ./gpu-setup-master-multi.sh
   ```

   Optionally, add the `--count=N` parameter where `N` indicates the desired number of GPU nodes to provision.

After you configure an elastic runtime environment with GPU nodes, select this runtime environment to run the classification engine training jobs in Metadata Command Center. For more information, see "Enable document classification" on page 21.

Enable document classification

After configuring an elastic runtime environment in Administrator, you must enable document classification and select the runtime environment to run the document classification training jobs in Metadata Command Center.

To enable document classification in your organization, perform the following steps:

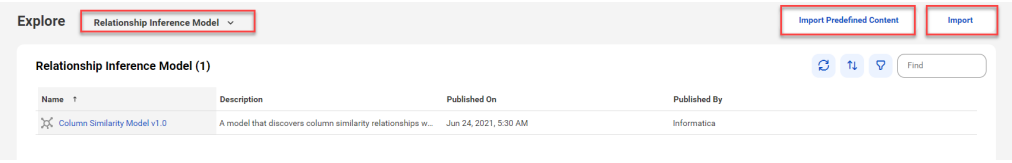
1. In Metadata Command Center, go to the **Configure** page.
2. Click the **Document Classification** tab.
3. Select **Enable Document Classification**.
4. Select the elastic runtime environment that you configured with GPU nodes in Administrator.
5. Click **Save**.

You can now enable the document classification capability for the catalog source to automatically identify and classify unstructured documents, such as PDFs and text files.

Import a relationship inference model

Import a relationship inference model if you want to configure the relationship discovery capability. You can either import a predefined relationship inference model, or import a model file from your local machine.

1. In Metadata Command Center, click **Explore** on the navigation panel.
2. Expand the menu and select **Relationship Inference Model**. The **Explore** page opens with the **Relationship Inference Model** menu.



3. Select one of the following options:
   - **Import Predefined Content**. Imports a predefined relationship inference model called Column Similarity Model v1.0.

- **Import**. Imports the predefined relationship inference model from your local machine. Select this if you previously imported predefined content into your local machine and the inference model is stored on the machine.
  To import a file, click **Choose File** in the **Import Relationship Inference Model** window and navigate to the model file on your local machine. You can also drag and drop the file.

The imported models appear in the list of relationship inference models on the **Relationship Discovery** tab.

CHAPTER 3

Create catalog sources in Metadata Command Center

Use Metadata Command Center to configure a catalog source for Amazon S3 and extract metadata.

When you configure a catalog source, you define the source system that you want to extract metadata from. Configure filters to include or exclude source system metadata before you run the job. Optionally, configure other capabilities, such as lineage discovery, autonomous enrichment, data profiling and quality, data classification, relationship discovery, and glossary association.

To provide stakeholders access to technical assets, you can assign access through stakeholder roles. You can also associate technical assets extracted from the catalog source to asset groups. If your catalog source references other source systems, you can create a connection assignment to the endpoint catalog source to view complete lineage.

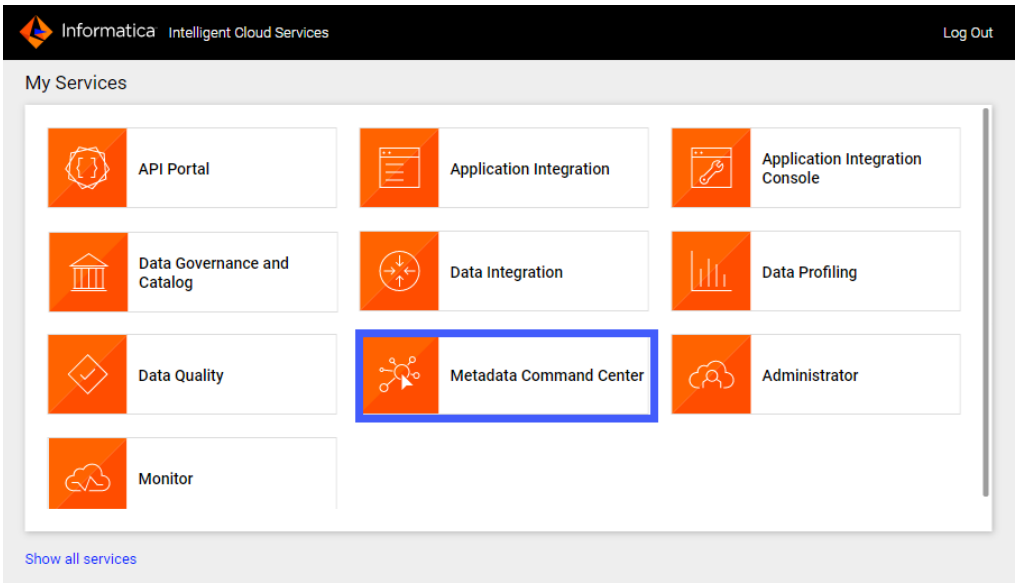
Step 1. Register a catalog source

When you register a catalog source, provide general information and connection values.

1. Log in to Intelligent Data Management Cloud.

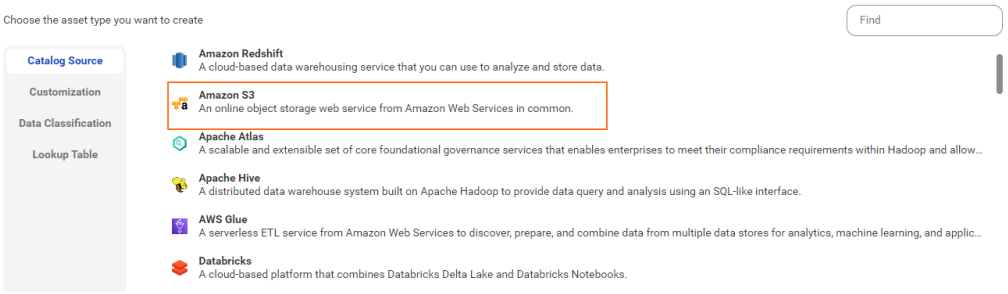
   The **My Services** page appears.
2. Click **Metadata Command Center**.

   The **Metadata Command Center** box appears on the **My Services** page.



The Metadata Command Center home page appears.

3. Click **New**.
4. Select **Catalog Source** from the list of asset types.
5. Select Amazon S3 from the list of catalog source types.



6. Click **Create**.

   The **New Catalog Source** page opens.
7. In the **General Information** section, enter a name and an optional description for the catalog source.

   Note: You can rename a catalog source after you create it, but to apply the change to all associated objects you must rerun the metadata extraction job.

   After you save the catalog source, you can update the description in Metadata Command Center and Data Governance and Catalog. The update appears only in the service in which you update it.
8. In the **Connection Information** area, select the connection that you created in Administrator.

   Note: To create or edit a catalog source, you need permissions on the connection to the source system. Select a connection that you have access to, or ask the administrator to grant the necessary permissions to the connection that you want to use.
9. Click **Connection Properties** to expand and view the connection properties for the selected connection.
10. Click **Test Connection** to test your connection to the source system.
11. Click **Next**.

The **Configuration** page appears.

Step 2. Configure capabilities

When you configure the Amazon S3 catalog source, you define the settings for the metadata extraction capability and other optional capabilities.

The metadata extraction capability extracts source metadata from external source systems. You can also configure other capabilities that the catalog source includes.

You can save the catalog source configuration at any point after you enter the connection information. After you save the catalog source, you can choose to run the catalog source job. To run the job once, click **Run**. To run metadata extraction and other capabilities on a recurring schedule, configure schedules on the **Schedule** tab.

Configure metadata extraction

When you configure the Amazon S3 catalog source, you choose a runtime environment, define filters, and enter configuration parameters for metadata extraction.

1. In the **Connection and Runtime** area, choose a serverless runtime environment or the Secure Agent group where you want to run catalog source jobs.

   Note: Serverless runtime environment options are available if the catalog source works with a serverless runtime environment.

2. Choose to retain, delete, or deprecate objects that are deleted from the source system in the catalog with the **Metadata Change Option**.

   - **Retain**. Retains objects that are deleted from the source system in the catalog. If you update or add a filter, the catalog retains objects extracted from the previous job and extracts additional objects that match the current filter. Objects deleted from the source system are not deleted from the catalog. Enrichments added on deleted objects and relationships are retained.
   - **Delete**. Deletes metadata from the catalog based on objects deleted from the source system and changes you make to the filter. Enrichments added on deleted objects and relationships are also permanently lost. Objects renamed in the source system are removed and recreated in the catalog.
   - **Deprecate**. When objects are deleted from the source system, the catalog changes the lifecycle of the imported objects to Obsolete. The lifecycle of objects also changes to Obsolete when you add exclusion filters or reduce the scope so that objects extracted in earlier jobs are no longer included. For example, the first job includes the sales schema. The second job also includes the sales schema, but excludes the sales.transactions table. In this case, the second job marks the sales.transactions object as Obsolete, even though the first job includes it. This does not impact enrichments applied to deprecated objects and relationships. Objects renamed in the source system are removed and recreated in the catalog. When you run the catalog source job again for other capabilities such as data classification, relationship discovery, or glossary association, the job doesn't consider obsolete objects. Obsolete objects remain in the catalog until they are purged when you run a **Purge Obsolete Objects** job on the **Explore** page.

   Note: You can also change the configured metadata change option when you run a catalog source.

3. In the **Filters** area, define one or more filter conditions to apply for metadata extraction.

   To define filters, you can either select an object type and enter the path to the object as the filter value, or select an object from a list of objects available in the source system.

   a. Select **Yes** to view filter options.

   b. From the Include/Exclude list, choose to include or exclude metadata based on the filter parameters.

c. Perform one of the following steps:

- From the Object type list, select an object type, depending on the object that you want to extract metadata from. Enter the path to the object as the filter value.
- In the filter value field, click the Search button and select an object from a list of objects available in the source system.
  The Object type field updates based on the selected object.

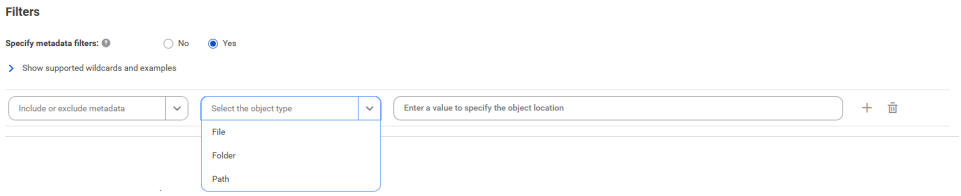
  If you select an object type and then click the Search button, the list of objects includes all object types, but you can only select objects that match the selected object type.
  You can edit the filter value after you select an object from the list.

  Note: You can only search for object types that work with the search functionality. If you don't see the Search button for the selected object, enter the object path as the filter value.

  Note: If the object metadata is available in Data Governance and Catalog, a check mark appears next to the object.

  Note: To select an object, you need to have permissions on the connection to the source system.

The following image shows the filter condition options:



d. To define an additional filter with an OR condition, click the **Add** icon.

4. In the **Configuration Parameters** area, enter the configuration parameters.

The following table describes the properties that you can enter in the **Catalog Source Configuration Options** section:

Property	Description
Extract Group Elements from Hierarchical Files	Select one of the following options to extract group or leaf elements from hierarchical files: - Yes. Extracts group elements from hierarchical files with the complete hierarchy of hierarchical fields. You can view the hierarchy of hierarchical files in the **Hierarchy** tab of assets in Data Governance and Catalog. - No. Extracts only leaf elements from hierarchical files without the complete hierarchy of hierarchical fields. You can extract group elements from hierarchical files for the following file types: - AVRO. Extracts and groups hierarchical files and hierarchical fields. - Parquet. Extracts and groups hierarchical files and hierarchical fields. - JSON. Extracts and groups hierarchical files and hierarchical fields. - XML. Extracts and groups elements and attributes. For XML file types, a maximum depth of 1000 elements is permitted within a single element in the hierarchy. - XSD. Extracts and groups elements and attributes. **Attention:** If you modify the **Extract Group Elements from Hierarchical Files** field and run the catalog source again, the asset page doesn't display the hierarchical elements in the correct hierarchy groups. If you modify the property value, purge the catalog source before you run it again.
Enable Extension-Based File Type Detection	Select one of the following options to detect file types by file extensions or by parsing the file contents: - Yes. Detects file types by file extensions. - No. Parses the file contents to detect file types. **Note:** You can detect file types by file extensions for the following file types: - CSV - TSV - TXT - XML
Use First Row as Header of Delimited Files	Select one of the following options to use the first row as the header or detect headers automatically for delimited files: - Yes. Detects column headers based on the following rules: - Duplicate headers get suffixed with '#' followed by a number, for example, ABC#1, ABC#2. The detection is not case-sensitive. - Empty column header values appear as UnknownColumn<position>, for example UnknownColumn2. - The header row in the file is detected even if it has a different number of columns than the data rows. - No. Detects headers automatically for delimited files.
Headers of Delimited Files	Specify values to determine headers of delimited files. Separate multiple values by commas. If any value from the list is found in the first row of the delimited file, then the first row is used as the header. **Note:** This parameter appears only if you choose No for the **Use First Row as Header of Delimited Files** parameter.
Treat Files Without Extension As	Select one of the following options to use to identify files without an extension: - Parquet - Avro - JSON

Property	Description
Enter File Delimiter	Specify the file delimiter if the file from which you extract metadata uses a delimiter other than the following list of delimiters: - Comma(,) - Horizontal tab (\t) - Semicolon (;) - Colon (:) - Pipe symbol (|) Enclose the delimiter in single quotes, such as '$'. Use a comma to separate multiple delimiters. **Note:** Adding a custom delimiter overrides the default list of delimiters. If you specify a delimiter, characters from the default list are not considered as delimiters.
Files to be excluded during partition discovery	Specify the regular expression of the files that you want to exclude during partition discovery. Use a comma to separate multiple regular expressions. For example, enter ".*json","Customer.csv","Parquet.*".
Treat Unrecognized Files As	Choose how to handle files with unrecognized formats during metadata extraction. Select one of the following options: - Structured files. Treats all files with unrecognized formats as structured files. - Unstructured files. Treats all files with unrecognized formats as unstructured files. **Note:** On a serverless runtime environment, unrecognized files are treated as structured files.
Unrecognized Files Override	Enter one or more file paths as exceptions to override whether the specific unrecognized files should be treated as structured or unstructured files. You can use the path-based overrides to apply different treatment rules to different files within the same catalog source.

The following table describes the properties that you can enter in the **Partitioned File Configuration** section:

Property	Description

Custom Partition Detection Configuration File	The configuration file in JSON format to detect custom partitions in the source system. The configuration file defines the pattern of the non-Hive style custom partitions.
	Note: The system automatically detects partitions if the date format of the partition key is in any of the following formats:
	- dd-MM-yy
	- dd-MM-yyyy
	- dd-MMM-yyyy
	For example, the system interprets dd-MM-yyyy as the partition format and extracts `Customer` as a hierarchical file instead of a folder in the following partition pattern:
	`Customer`
	`|__01-01-2022`
	`|  |  |__ dataFile1.parquet`
	`|  |  |__ dataFile2.parquet`
	`|  |  |__ ...`
	`|__02-02-2022`
	`|__...`
	The system doesn't automatically detect partitions with other patterns, such as MM-dd-yyyy. If the pattern is different, you need to define it in JSON format. For example, `{"CustomPartitionPatterns": ["MM-dd-yyyy"]}`
	To detect the epoch time format, define it in JSON format as: `{"CustomPartitionPatterns": ["@"]}`

Partitioned Pruning Configuration Options	The configuration file in JSON format for partition pruning. The configuration file contains additional information to identify partitions and determine the relevant schema in the source system.
Enable Partition Detection	Specify if you want to enable partition detection.
Enable Partition Pruning	Specify if you want to enable partition pruning. Default is Yes. **Note:** Applicable if you enable partition detection. If you run the catalog source job to extract incremental metadata with partition pruning, the job considers the pruned partitioned data that is changed or new since the last job run.

5. Configure additional capabilities for the catalog source by clicking on the tabs.

Filter guidelines and examples

You can add metadata extraction filters when you configure the catalog source. To create a filter, you can use choose from file names, folder names, or paths.

Consider the following rules and guidelines when you enter filter values:

- Filters are case-sensitive.
- Use an asterisk to represent multiple characters in a folder name, file name, and a single folder level in a folder hierarchy. For example, A* matches A, Ab, ABC.
- For file filters, specify only the file name.

- If a file name contains an asterisk, the filter considers it as a wildcard and not a special character. To ignore an asterisk as a wildcard, enclose it in double quotes (") in the filter.
- Use a forward slash as a separator in path hierarchies. You can add a path in folder and path filters.
- Use an asterisk as a path placeholder in folder and path filters. For example, folder1/*/folder3. Here, the filter includes all folders under folder1.
- Use two asterisks to indicate zero or more levels of folders in folder and path filters. The pattern with two asterisks is recursive. The processing time is longer as the data volume increases.

Important: It is recommended that you either use only a path filter or use a combination of a folder and a file filter.

Examples

You can include or exclude metadata from folders, files, or paths.

Folder filters

Folder filters apply to folders included in the source system.

For example:

- To include or exclude metadata from 'Folder2' located inside 'Folder1', select **Folder** as the object type and enter `Folder1/Folder2` in the value field.
- To include or exclude metadata from 'Folder2' located in any folder under 'Folder1', select **Folder** as the object type and enter `Folder1/*/Folder2` in the value field.
- To include or exclude metadata from 'Folder2' located two levels under 'Folder1', select **Folder** as the object type and enter `Folder1/*/*/Folder2` in the value field.
- To include or exclude metadata from 'Folder2' located at any level under 'Folder1', select **Folder** as the object type and enter `Folder1/**/Folder2` in the value field. This is a recursive search, and therefore the processing time can be longer.

File filters

File filters apply to the files included in folders that you filter. The file filter is recursive. If you don't provide any folder filters, the file filters apply to the entire folder hierarchy.

For example:

- To include or exclude metadata from all files with the name 'File1.csv' located in the source directory, select **File** as the object type and enter `File1.csv` in the value field. Metadata Command Center recursively searches for files that match the filter criteria in all folders in the source directory.
- To include or exclude metadata from all files with names that start with 'File' and end with 'ame.csv', select **File** as the object type and enter `File*ame.csv` in the value field.
- To include or exclude metadata from all files with names that end with 'File.csv', select **File** as the object type and enter `*File.csv` in the value field.
- To include or exclude metadata from all files with the name 'File' and files that start with the name 'File' followed by one or more characters, select **File** as the object type and enter `File*` in the value field.
- To include or exclude metadata from all files with names that contain the word 'File', select **File** as the object type and enter `*File*` in the value field.
- To include or exclude metadata from all files with the name 'Fi*le.csv', select **File** as the object type and enter `Fi"*"le.csv` in the value field.

Path filters

Path filters apply to the files and folders in the path that you filter. The path filter is non-recursive. If you provide only the file or folder names, the path filters apply to the first level files or directories.

For example:

- To include or exclude metadata from files and folders with names that start with 'Item1' in the first level directory, select **Path** as the object type and enter `Item1*` in the value field.
- To include or exclude metadata from the 'File1' file in the 'Folder1' folder, select **Path** as the object type and enter `Folder1/File1` in the value field.
- To include or exclude metadata from files or folders with names that contain the word ‘Subfolder’ in the ‘Folder1’ folder, select **Path** as the object type and enter `Folder1/*Subfolder*` in the value field.
- To include or exclude metadata from files or folders with the name 'File1' in any subfolder of the 'Folder1' folder, select **Path** as the object type and enter `Folder1/*/File1` in the value field.
- To include or exclude metadata from all files and subfolders in the 'Folder1' folder, select **Path** as the object type and enter `Folder1/*` in the value field.
- To include or exclude metadata from files or folders with the name ‘File1’ located at any level in the ‘Folder1’ folder, select **Path** as the object type and enter `Folder1/**/File1` in the value field. This is a recursive search, and therefore the processing time can be longer.

Configure data profiling and quality

Enable the data profiling capability to evaluate the quality of metadata extracted from the Amazon S3 source system.

1. Click the **Data Profiling and Quality** tab.
2. Expand **Data Profiling** and select **Enable Data Profiling**.

   Note: Ensure that you have permissions on all the staging connections that you use in your data profiling configuration. You can't run the job if you don't have permissions on the connections that you use. Select connections that you have access to, or ask the administrator to grant the necessary permissions on the connections that you want to use.
3. In the **Connection and Runtime** area, choose the Secure Agent group where you want to run catalog source jobs.
4. Optionally, in the **Filters** area, specify additional filters in addition to metadata filters:
   a. Select **Yes** to view filter options.
   b. From the Include/Exclude list, choose to include or exclude metadata based on the filter parameters.
   c. From the Object type list, select **File** or **Folder**.
   d. Enter the path to the object as the filter value.

      Filters can contain the following wildcards:

      - Question mark. Represents a single character.
      - Asterisk. Represents multiple characters.

      Examples:

      - File: You extracted metadata from a folder that contains multiple files and now you want to run a profile on a specific file. Select **File** from the Object type option and select **Named** from the Filter Criteria option, and then enter the file name in the input field.

- Folder: You extracted metadata from a folder that contains multiple folders and now you want to run a profile on a specific folder. Select **Folder** from the Object type option and select **Named** from the Filter Criteria option, and then enter the folder name in the input field.

To include or exclude multiple objects, click the **Add** icon to add filters with the AND condition.

5. In the **Parameters** area, configure the parameters.

   The following table describes the parameters that you can enter:

Parameter	Description
Modes of Run	Determine the type of data that you want the data profiling task to collect. Choose one of the following options: • Keep signatures only. Collects only aggregate information such as data types, average, standard deviation, and patterns. • Keep signatures and values. Collects both signatures and data values.
Profiling Scope	Determine whether you want to run data profiling only on the changes made to the source system or on the entire source system. Choose one of the following options: • Incremental. Includes only source metadata that is changed or updated since the last profile run. • Full. Includes the entire metadata that is extracted based on the filters applied for extraction.
Sampling Type	You can run the data profiling task on all rows for an Amazon S3 source system.
Elastic Runtime Environment	Select an elastic runtime environment for complex file types, including Avro and Parquet.
Staging Connection	The staging connection where data profiling results are stored temporarily during the execution. Applicable for data profiling on complex file types.
Maximum Precision of String Fields	The maximum precision value for profiles on string data type.
Text Qualifier	The character that defines string boundaries. If you select a quote character, profiling ignores delimiters within the quotes. Select a qualifier from the list. **Note:** Default is Double Quote.

Parameter	Description
Code Page for Delimited Files	Select a code page that the Secure Agent can use to read and write data. Use this option to ensure that profile results for assets with non-English characters don't include junk characters. Default value is UTF-8. Choose one of the following options: • MS Windows Latin 1. Select for ISO 8859-1 Western European characters. • UTF-8. Select for Unicode and non-Unicode characters. • Shift-JIS. Select for double-byte characters. • ISO 8859-15 Latin 9 (Western European). • ISO 8859-2 Eastern European. • ISO 8859-3 Southeast European. • ISO 8859-5 Cyrillic. • ISO 8859-9 Latin 5 (Turkish). • IBM EBCDIC International Latin-1.
Escape character for delimited files	You can specify an escape character if you need to override the default escape character. An escape character ignores a delimiter character in an unquoted string if the delimiter is part of the string value. If you specify an escape character, the data profiling task overrides the default escape character that the Metadata Extraction job detects and considers the specified escape character. It then reads the delimiter character as a part of the string value. If you don't specify an escape character, the data profiling task considers the default escape character that the Metadata Extraction job detects and reads the delimiter character as a part of the string value.
Read Multiple Line JSON Files	By default, the data profiling job reads each JSON schema as a single line. Select Yes to read input that spans across multiple lines.

6. Expand **Data Quality** and select **Enable Data Quality**.

   Note: You can click **Use Data Profiling Parameters** to use the same parameters as in the **Data Profiling** section.

   Note: Ensure that you have permissions on all the staging and flat file connections that you use in your data quality configuration. You can't run the job if you don't have permissions on the connections that you use. Select connections that you have access to, or ask the administrator to grant the necessary permissions on the connections that you want to use.

7. In the **Connection and Runtime** area, choose the Secure Agent group where you want to run catalog source jobs.
8. In the **Parameters** area, configure the parameters.

The following table describes the properties that you can enter:

Parameter	Description
Data Quality Rule Automation	Enable the option to automatically create or update rule occurrences for data elements in the catalog source. Choose one of the following options: • **Apply on Data Elements linked with Business Dataset**. Creates rule occurrences for all data elements that are linked with business data sets in the catalog source. • **Apply on all Data Elements**. Creates rule occurrences for all data elements in the catalog source.
Data Quality Remediation	Enable the option to specify a flat file connection to store the list of failed rows so that users can remediate poor data quality scores. Choose one of the following options: • **No**. Doesn't enable the data quality failure ticket option. • **Yes**. Shows a list of flat file connections where you write failed rows to customer-managed locations.
Data Quality Failure Ticket	Specify whether you want to create data quality failure tickets for poor data quality scores based on the threshold defined for the rule occurrence in Data Governance and Catalog. Choose one of the following options: • **No**. Doesn't automatically create data quality failure tickets when the data quality scores are poor. • **Yes**. Automatically creates data quality failure tickets based on the data quality threshold values you define in Data Governance and Catalog, and notifies you when a data quality score is below the threshold. **Note:** You must configure a workflow event for the data quality failure and enable the event in Metadata Command Center.
Cache Result	Specify how you want to preview rule occurrence results. Select **Agent Cache** if you want to generate a cache file in the runtime environment and to preview the cached results faster in subsequent data preview runs. The results are cached for seven days by default after the first run in the runtime environment. Select **No Cache** if you don't want to cache the preview results and view the live results.
Run Rule Occurrence Frequency	Specify whether you want to run data quality rules based on the frequency defined for the rule occurrence in Data Governance and Catalog.
Sampling Type	You can run the data quality task on all rows for an Amazon S3 source system.
Elastic Runtime Environment	Select an elastic runtime environment for complex file types, including Avro and Parquet.
Staging Connection	The staging connection where data quality results are stored temporarily during the execution. Applicable for data profiling on complex file types.

Parameter	Description
Maximum Precision of String Fields	The maximum precision value for profiles on string data type.
Text Qualifier	The character that defines string boundaries. If you select a quote character, profiling ignores delimiters within the quotes. Select a qualifier from the list. **Note:** Default is Double Quote.
Code page for delimited files	Select a code page that the Secure Agent can use to read and write data. Use this option to ensure that rule results for assets with non-English characters don't include junk characters. Default value is UTF-8. Choose one of the following options: • MS Windows Latin 1. Select for ISO 8859-1 Western European characters. • UTF-8. Select for Unicode and non-Unicode characters. • Shift-JIS. Select for double-byte characters. • ISO 8859-15 Latin 9 (Western European). • ISO 8859-2 Eastern European. • ISO 8859-3 Southeast European. • ISO 8859-5 Cyrillic. • ISO 8859-9 Latin 5 (Turkish). • IBM EBCDIC International Latin-1.
Escape character for delimited files	You can specify an escape character if you need to override the default escape character. An escape character ignores a delimiter character in an unquoted string if the delimiter is part of the string value. If you specify an escape character, the data quality task overrides the default escape character that the Metadata Extraction job detects and considers the specified escape character. It then reads the delimiter character as a part of the string value. If you don't specify an escape character, the task considers the default escape character that the Metadata Extraction job detects and reads the delimiter character as a part of the string value.
Read Multiple Line JSON Files	By default, the data quality job reads each JSON schema as a single line. Select Yes to read input that spans across multiple lines.

Configure data observability

Enable the data observability capability to receive anomaly notifications for metadata extracted from the source system.

To include profiling anomalies, enable data profiling for the catalog source. For accurate results, set the **Profiling Scope** option to **Full** in the **Data Profiling and Quality** configuration.

1. Click the **Data Observability** tab.
2. Select **Enable Data Observability**.
3. In the **Parameters** area, enter a value to specify the minimum number of catalog source job runs required to start detecting anomalies. For example, if you enter a value of 5, then the job starts to detect anomalies from the sixth run.

   Note: This option does not apply to anomalies that require a minimum of two runs to generate.
4. Specify the maximum number of anomaly events to generate for each catalog source run. Default is 1000.

5. Optionally, in the **Asset Filters** area, enable filters to run data observability on specific assets.

   Specify data profiling filters if you want to override the data profiling filter and extract profiling anomalies for assets based on a different filter.

6. To add a data profiling filter, perform the following steps:
   a. Select **Yes** to view filter options.
   b. From the **Include/Exclude** list, choose to include or exclude metadata.
   c. From the **Specify data profiling filters** list, select an object type.
   d. Enter a value to specify the object location.

      For example, you ran a profile on multiple schemas, and now you want to generate profiling anomalies on a specific schema. Enter the schema name in the input field.

   e. To include or exclude multiple objects, click the **Add** icon to add filters with the OR condition.

7. Optionally, in the **Metric Filter** area, specify filters to generate events based on specific metrics. If you don't include a filter, the job detects both profiling and other anomalies applicable to the catalog source.
   a. Select **Filter conditions**.
   b. From the **Include Metric** list, choose to include or exclude a metric.
   c. From the **All Metrics** list, select a metric for which you want to generate events.
   d. From the **Sensitivity** list, select the sensitivity of the anomaly.
   e. From the **Select detection rules** list, select rules that you apply to detect anomalies.

      The following table describes the detection rules that you can select:

Option	Description
Static Data	Detects the following anomalies: - Percentage variation. If percentage values other than 0% or 100% remain constant for three or more profile runs, any subsequent change generates a Percentage Variation anomaly. - Count variation. A change in a value that is usually constant indicates that there is an anomaly.
100% or 0% Change Detection	Detects the following anomalies: - Drop from maximum. A drop from 100% to a lower value generates a Drop from Maximum anomaly. - Surge from minimum. A spike from 0% to any higher value generates a Surge from Minimum anomaly.

Option	Description
Standard Deviation	Detects the following anomalies: - Drop in transition. A drop from a value that has been decreasing at a constant rate to a significantly lower value generates a Drop in Transition anomaly. - Surge in transition. A spike from a value that has been increasing at a constant rate to a significantly higher value generates a Surge in Transition anomaly. - Deviation. If the percentage values of a record change with each profile run, the algorithm calculates an expected range based on the changes observed in each profile run. A new value that falls outside of this expected range generates a Deviation anomaly.
Breaking Trends	Detect the following types of count-based anomalies: - Drop. A drop in a value that has been increasing to a lower value generates a Drop anomaly. - Surge. A spike in a value that has been decreasing to a higher value generates a Surge anomaly.

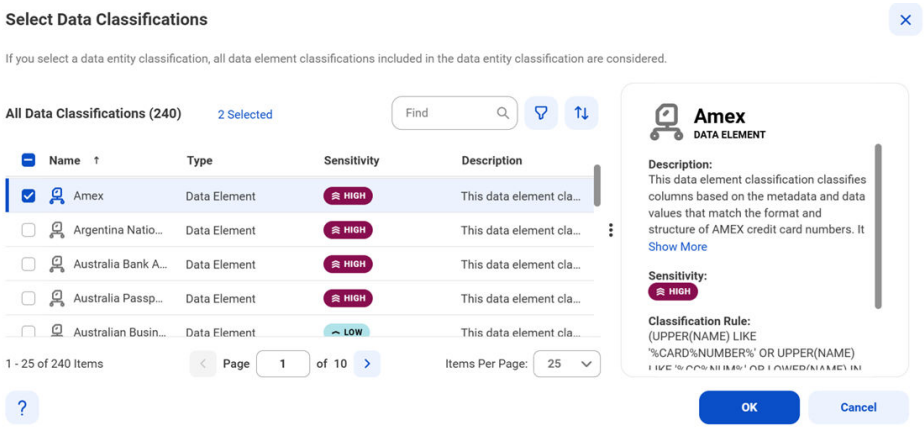
Configure data classification

Enable the data classification capability to identify and organize data into relevant categories based on the functional meaning of the data.

Before you enable document classification for the catalog source, perform the following tasks:

- Configure an elastic runtime environment with GPU nodes. For more information, see "Configure runtime environment for document classification" on page 20.
- Select the runtime environment to run the document classification training jobs. For more information, see "Enable document classification" on page 21.

1. Click the **Data Classification** tab.
2. Select **Enable Data Classification**.
3. Choose one or both of the following options:
   - **Generated Data Classifications**. CLAIRE automatically generates data classifications for the data elements.
   - **Data Classification Rules**. Choose from predefined or custom data classifications.
     1. Click **Add Data Classification**. The **Select Data Classifications** dialog box appears.

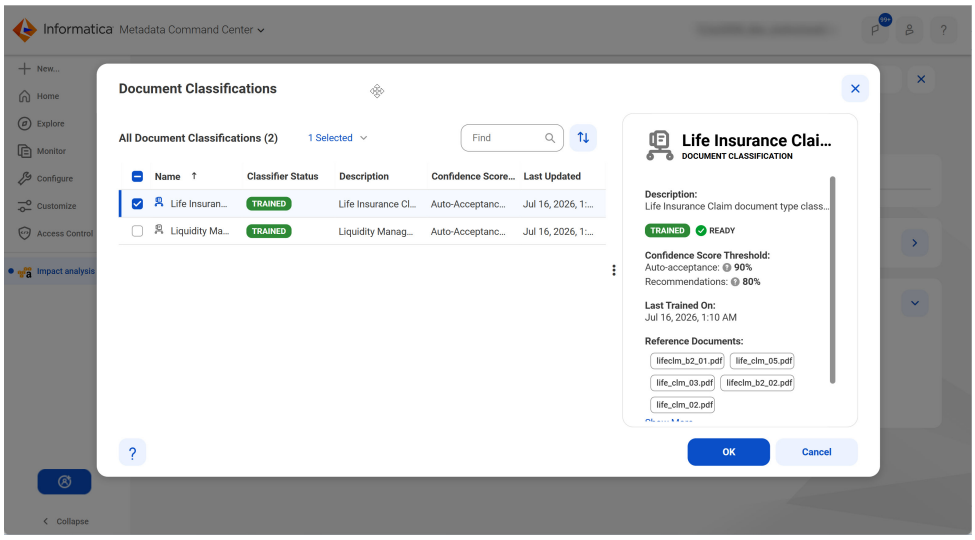


     2. Select the data classifications that you want to use.

3. Click **OK**.

4. Optionally, select **Enable Document Classification** to automatically identify and classify unstructured documents, such as PDFs and text files, for enhanced search and discovery.

   a. Click **Select Document Classifications**. The **Document Classifications** dialog box appears.



   b. Choose one or more document classifications that you created for your organization.

   c. Click **OK**.

Configure relationship discovery

Enable the relationship discovery capability to identify pairs of similar columns and relationships between tables within a catalog source.

Before you configure relationship discovery, perform the following tasks:

- Import a relationship inference model. For more information about importing a relationship inference model, see “ Import a relationship inference model” on page 21.
- Enable data profiling on the **Data Profiling and Quality** tab, and select **Keep Signatures and Values** as the run mode in the **Parameters** section. These configurations enable you to retain values of the columns in the profiling results and discover relationships.

1. Click the **Relationship Discovery** tab.
2. Select **Enable Relationship Discovery**.
3. In the **Column Similarity** area, select the **Relationship Inference Model**.

   Note: The relationship inference models that you imported appear in the **Relationship Inference Model** field.

4. In the **Joinable Tables Relationship** area, specify the **Containment Score Threshold** to identify joinable table relationships within the catalog source. This score is an indicator of the data overlap between any two given columns which determines whether the tables are joinable.

   Note: A higher score means that the objects have more overlapping data and a lower score means lesser overlapping data between the two objects. A containment score threshold lower than 0.4 might result in a large number of false positives.

After you run the catalog source job, you can view the inferred relationships on the **Relationships** tab of the extracted assets in Data Governance and Catalog.

Configure glossary association

Enable the glossary association capability to associate glossary terms with technical assets, or to get recommendations for glossary terms that you can manually associate with technical assets in Data Governance and Catalog.

Metadata Command Center considers all published business terms in the glossary while making recommendations to associate your technical assets.

1. Click the **Glossary Association** tab.
2. Select **Enable Glossary Association**.
3. Select **Enable auto-acceptance** to automatically accept glossary association recommendations.
4. Specify the **Confidence Score Threshold for Auto-Acceptance** to set a threshold limit based on which the glossary association capability automatically accepts the recommended glossary terms.

   Note: Specify a percentage from 80 to 100. If the score is higher than the specified limit, the glossary association capability automatically assigns a matching glossary term to the data element.
5. Select **Enable Below-threshold Recommendations** to receive glossary association recommendations below the auto-acceptance threshold. If you enable auto-acceptance, you can enable below-threshold recommendations to receive glossary recommendations below the auto-acceptance threshold.
6. Specify the **Confidence Score Threshold for Recommendations** to set a threshold based on which the glossary association capability makes recommendations

   If you enable auto-acceptance, specify a percentage from 80 to the selected auto-acceptance threshold. You can accept or reject the recommended glossary terms that fall within this range in Data Governance and Catalog.

   If you disable auto-acceptance, specify a percentage from 80 to 100 inclusive.
7. Choose to automatically assign business names and descriptions to technical assets. You can then choose to retain existing assignments and only assign business names and descriptions to assets that don't have assignments, or allow overwrite of existing assignments.

   By default, existing assignments are retained.
8. Optional. Choose to ignore specific parts of data elements when making recommendations. Select **Yes** and enter prefix and suffix keyword values as needed.

   Click **Select** to enter a keyword. You can enter multiple unique prefix and suffix keywords. Keyword values are case insensitive.
9. Optional. Choose specific top-level business glossary assets to associate with technical assets. Selecting a top-level asset selects its child assets as well. Select **Top-level Glossary Assets** and specify the assets on the **Select Assets** page.
10. Optional. Choose to use abbreviations and synonym definitions from lookup tables for accurate glossary association. Select **Yes** to enable, and then click **Select** to upload a lookup table.
11. Click **Next**.

    The **Associations** page appears.

Configure automation

Enable lineage-based propagation to receive enrichments propagated from catalog sources connected through lineage.

Enable Lineage-based Propagation on the Autonomous Enrichment tab of the Configure page before you configure this catalog source to receive enrichments. For more information, see *Administration*.

With the lineage-based propagation settings on this tab, you configure how this catalog source receives enrichments propagated from connected catalog sources. Enrichments propagate only to assets that don't already have enrichments.

1. Click the **Automation** tab.
2. In the **Autonomous Enrichment** section, select **Enable Lineage-based Propagation**.

   The lineage-based propagation settings appear.
3. Select the enrichment types to receive from connected catalog sources.

   You can select Classification, Description, Business Name, and Glossary Term.
4. Choose how to apply received enrichments.
   - Select **Auto-accepted** to apply received enrichments automatically.
   - Select **Recommended** to list them as suggestions that you can curate in Data Governance and Catalog.
5. Choose whether to receive enrichments only from certified assets.

   Assets are certified in Data Governance and Catalog. This option is selected by default. If you clear it, this catalog source also receives enrichments from assets that are not certified, which might be less reliable.
6. Choose where this catalog source receives enrichments from.
   - Select **Catalog Source** to receive enrichments from a specific catalog source.
   - Select **Nearest Neighbor** to receive enrichments from the closest connected catalog source based on lineage.
7. If you select **Catalog Source**, select a catalog source from the list.

   Ensure that the catalog source you select is within 15 lineage hops. Enrichments don't propagate from a catalog source beyond that distance.
8. If you select **Nearest Neighbor**, configure the lineage settings:
   a. Set **Lineage Direction** to search upstream, downstream, or in both directions.

      Upstream catalog sources feed data into this catalog source. Downstream catalog sources receive data from this catalog source. Select both directions to search upstream and downstream at the same time.

   b. Set **Number of Hops** to define how far along lineage to search.

      The search stops at the first connected catalog source that has enrichments, up to a maximum of 15 hops.
9. Click **Save**.

When you save, the catalog source begins receiving enrichments created from that point onward. Saving doesn't retrieve enrichments created prior to this date and time. To receive such enrichments, run the catalog source job manually and include the Autonomous Enrichment capability.

Step 3. Associate stakeholders and asset groups

Associate users or user groups within a stakeholder role as stakeholders for technical assets in Data Governance and Catalog. Also, you can choose to assign technical assets extracted from the catalog source

to asset groups. You can then use access policies to control permissions on assets that are assigned to asset groups.

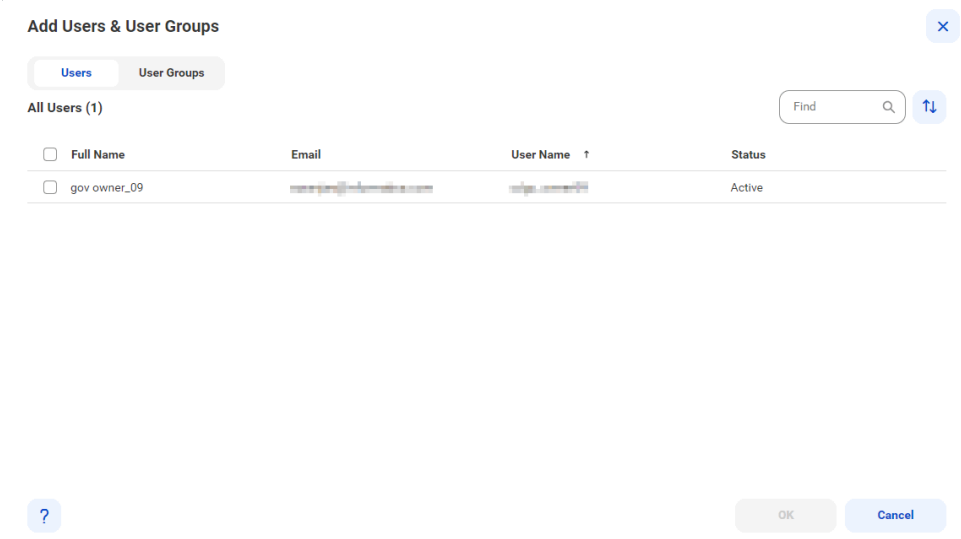
Verify that the administrator assigned users and user groups to the stakeholder role that you want to associate with technical assets.

1. To associate users or user groups as stakeholders with technical assets extracted from the catalog source, perform the following steps:
   a. On the **Associations** page, click **Stakeholders**.
   b. Select **Assign Stakeholders**.
   c. Select **Propagate to Child Assets** to propagate stakeholders of a parent asset to child assets without stakeholders.
   d. Select **Append Stakeholders** to propagate stakeholders of a parent asset to all child assets including assets with stakeholders assigned.

      Note: To enable **Append Stakeholders**, you must enable **Propagate to Child Assets**.

      You can also override the propagation settings from the catalog source page in Data Governance and Catalog. For more information about stakeholder propagation, see *Stakeholder Propagation* in *Administration*.
   e. Select a stakeholder role.
   f. Click **Select** to add users and user groups from the stakeholder role as stakeholders for the technical assets.

      The **Add Users & User Groups** dialog box displays a list of users and user groups assigned to the selected stakeholder role.



   g. Select one or more users or user groups to assign as stakeholders for the technical assets, and click **OK**.

      Only the selected users and user groups belonging to the specified stakeholder role are granted the permissions to technical assets.
   h. To assign users or user groups from another stakeholder role, click **Add** and then repeat the steps.
2. To assign asset groups to technical assets extracted from the catalog source, perform the following steps:
   a. On the **Associations** page, click **Asset Groups**.

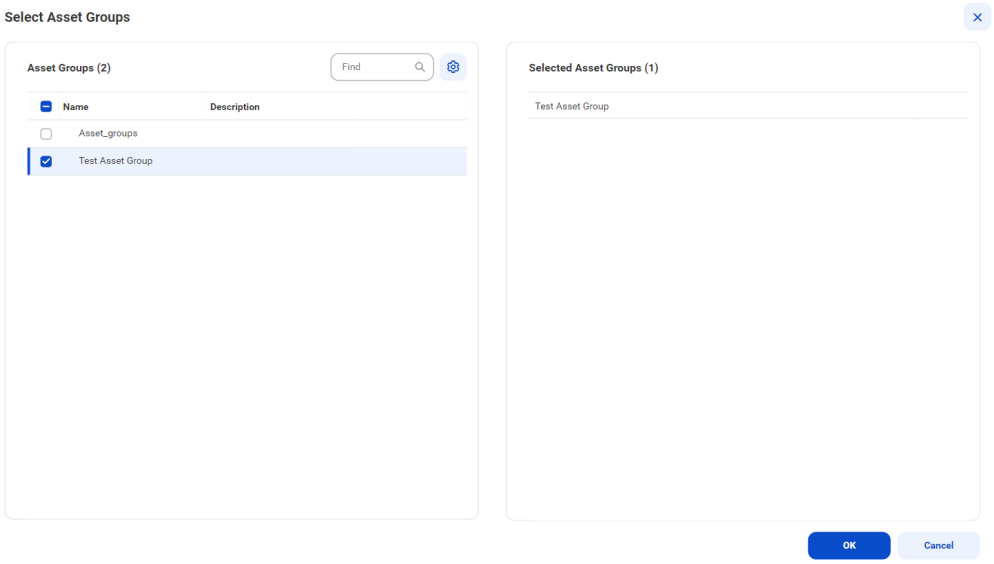
b. Select **Assign Asset Groups**.

c. Click **Select**.

The **Select Asset Groups** dialog box displays the list of asset groups.

If you enabled an access policy that includes an asset group, you can only view assets that belong to that asset group.

3. Select the asset groups to which you want to assign technical assets extracted from the catalog source, and click **OK**.



4. Choose to save and run the job or to schedule a recurring job.
   - To save and run the job, click **Save** and then **Run**.
   - To schedule a recurring job, click **Next** to open the **Schedule** page.

Step 4. Run or schedule the job

Choose to run a catalog source job manually, or configure it to run on schedule.

Note: You can't run multiple jobs simultaneously.

You can choose to perform a full or an incremental metadata extraction. A full metadata extraction extracts all objects from the source to the catalog. An incremental metadata extraction extracts only the changed and new objects since the last successful catalog source job run. Incremental metadata extraction doesn't remove deleted objects from the catalog and doesn't extract metadata of code-based objects if applicable.

When you run an incremental metadata extraction job with a filter to include metadata from objects, the job extracts only the objects that have the latest timestamp since the last successful job.

Note: If incremental metadata extraction is available for the catalog source, the option is always visible, but incremental metadata extraction jobs can run only after a scheduled full metadata extraction job. Manual jobs do not enable the incremental metadata extraction option.

Run the job manually

Click **Save** to save the catalog source and click **Run**. On the **Run Catalog Source Job** window, click **Run** to run the job.

You can override the capabilities that you selected while configuring your catalog source on the **Configuration** page. The first time you run the catalog source job, the metadata extraction capability is mandatory. From the second run onwards, you can choose to override the configured metadata change option. You can retain, delete, or deprecate objects that are deleted from the source in the catalog. For subsequent runs of the catalog source job, the metadata extraction capability is optional.

Note: You can choose incremental metadata extraction for subsequent runs only after one full metadata extraction job completes successfully. Incremental metadata extraction jobs run with the **Retain** metadata change option even if you set the option to **Delete** or **Deprecate** in the catalog source.

Note: To run a catalog source job, you need permissions on the connection to the source system. To run a catalog source job for catalog sources that reference other source systems, you need permissions on the connections for all the reference source systems.

Run the job on a schedule

You can choose to run metadata extraction and other capabilities on a recurring schedule. You can't choose incremental metadata extraction and full metadata extraction in the same schedule. To create a schedule for incremental metadata extraction, you must have completed at least one full metadata extraction job successfully. If not, first create a schedule for a full metadata extraction.

If an incremental metadata extraction is scheduled to run when the last run details aren't available, the job first performs a full metadata extraction, followed by incremental metadata extraction on subsequent runs.

For example, this can happen in the following scenarios:

- You create schedules for both incremental metadata extraction and full metadata extraction, but schedule the incremental extraction to run before the first full metadata extraction job.
- You create schedules for both incremental metadata extraction and full metadata extraction, but delete the full metadata extraction schedule before its first run.

1. On the **Schedule** tab, select **Run on Schedule**.
   The **Schedule** configuration page opens.
2. Click the checkbox corresponding to each capability that you want to include in the schedule.
3. Enter the start date, time zone, and the interval at which you want to run the job.
4. You can manage additional schedules using the following options:
   - To create a new schedule, click the **Add** button.
   - To delete a schedule, click the **Delete** button.
   - To enable or disable a schedule, click the **Enable Schedule** toggle button.

   Note: You can create a maximum of one schedule per capability that you enable. If you purged a catalog source or did not run the metadata extraction job, the catalog source job runs metadata extraction before running other scheduled capabilities.

   Note: To create a schedule, you need permissions on the connection to the source system. If you lose permissions on the connection after you create a schedule, the scheduled jobs continue to run.
5. Click **Save** to save the schedule.

Monitor job status

After the job runs, you can monitor the status of the job on the **Overview** page of the job.

For more information about job monitoring, see *Administration*.

CHAPTER 4

View results in Data Governance and Catalog

After Metadata Command Center runs a job, you can view the results in Data Governance and Catalog where the catalog source and its elements are called technical assets. You can view a catalog source as a hierarchy. Expand each technical asset to see its components.

When referenced source systems are connected to a catalog source, you can expand the hierarchy to see details about the technical asset's component elements.

You can view the data lineage of an asset contained within a catalog source to see individual elements such as data sources, calculations, and filters. When you view data lineage, you can see the individual upstream elements that contribute data or expressions to each component of a data flow or catalog source.

View metadata extraction results

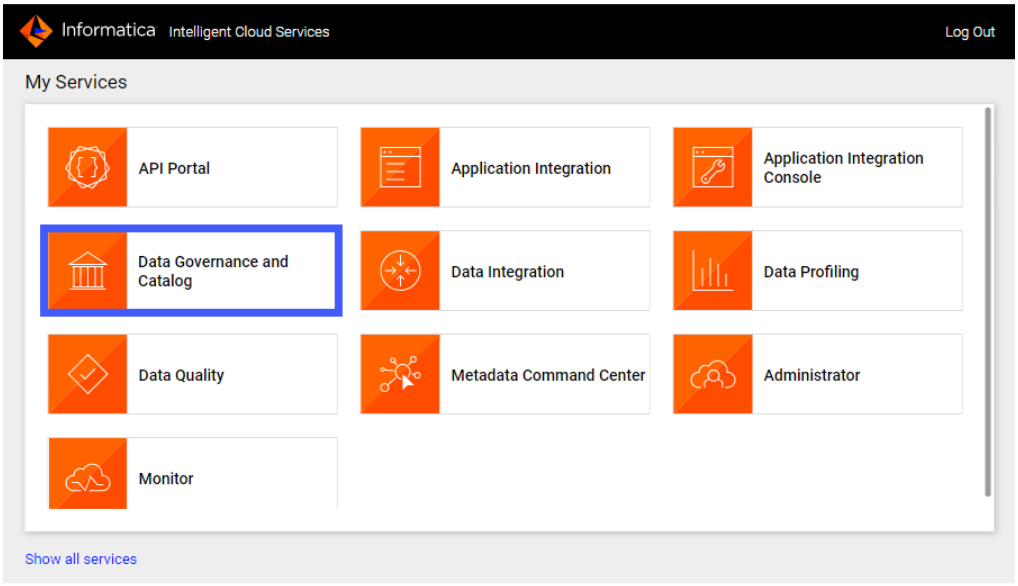
After a job runs in Metadata Command Center, view the results in Data Governance and Catalog. You can view details about source system contents in a hierarchical structure and trace data lineage.

1. Log in to IDMC.

   The **My Services** page appears.

2. Click Data Governance and Catalog.

   The following image shows the Data Governance and Catalog box on the **My Services** page:



3. On the Data Governance and Catalog home page, click the number in the **Technical Assets** panel.

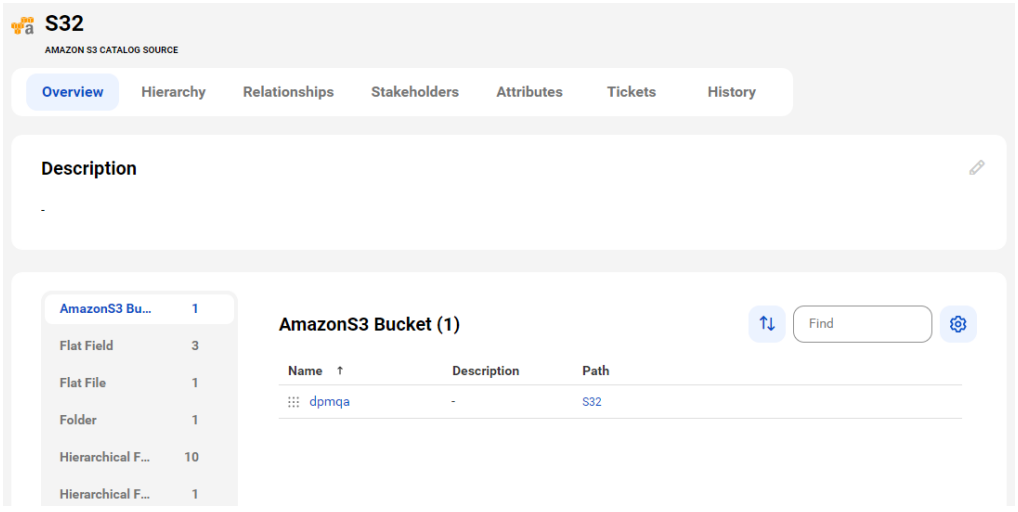
   The **Technical Assets** page opens.

4. Select **Catalog Source** in the **Filter** list.

   The list of catalog sources opens.

5. Search for the catalog source from which you extracted metadata, and click the name.

   The **Overview** tab of the asset opens.



Note: If a new record is added to a file in the Amazon S3 source system, the timestamp of the file gets updated. If you then run an incremental metadata extraction job, the file gets extracted because objects are extracted based on the updated timestamp.

6. View the asset from different perspectives by clicking on the tabs.

   Note: If an incremental metadata extraction job includes a view, it extracts the corresponding tables even if the tables were extracted in previous runs.

For more information about working with assets, see Working with Assets in *Data Governance and Catalog* help.

View data lineage

Data lineage is a visual representation of the flow of data across the systems in your organization. Lineage depicts how the data flows from the system of its origin to the system of its destination.

Data lineage views are available for technical assets in the catalog source. You can view lineage at the catalog source, data set, or data element level.

The lineage at the catalog source level shows how data flows from one catalog source to another. The lineage at the data set and the data element levels show how other technical assets such as files or tables contribute to the selected asset.

Note: File system assets can be source and target endpoint objects for other catalog sources, for example Business Intelligence or ETL sources. File system sources don't create any references to external assets. If you didn't assign them to any reference assets created by other catalog sources, the **Lineage** tab doesn't display the data flow.

If linking catalog sources is available for your catalog source, you can use Metadata Command Center to generate data lineage based on rules or by generating automated lineage with CLAIRE. You can choose source and target catalog sources and objects to link and generate lineage.

To determine whether linking catalog sources is available for your catalog source, navigate to the **Configuration** tab of the **Link Catalog Sources** page. The catalog source must appear in the list of source and target catalog sources.

For information about linking catalog sources, see *Link catalog sources* in the Administration help.

View lineage at the catalog source level

The catalog source level shows how data flows from one catalog source to another with the lineage aggregating data from the data set and data element levels.

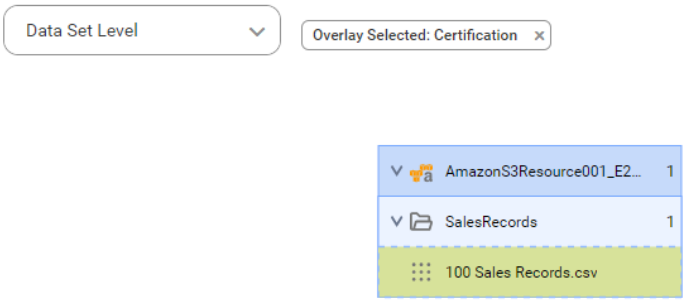
To view data lineage at the catalog source level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Catalog Source Level**.

View lineage at the data set level

The data set level displays individual sets of data in the data flow.

To view lineage at the data set level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Data Set Level**.

The following image shows the data set level lineage for the 100 Sales Records.csv file :



View lineage at the data element level

The data element level displays elements of data sets in the data flow.

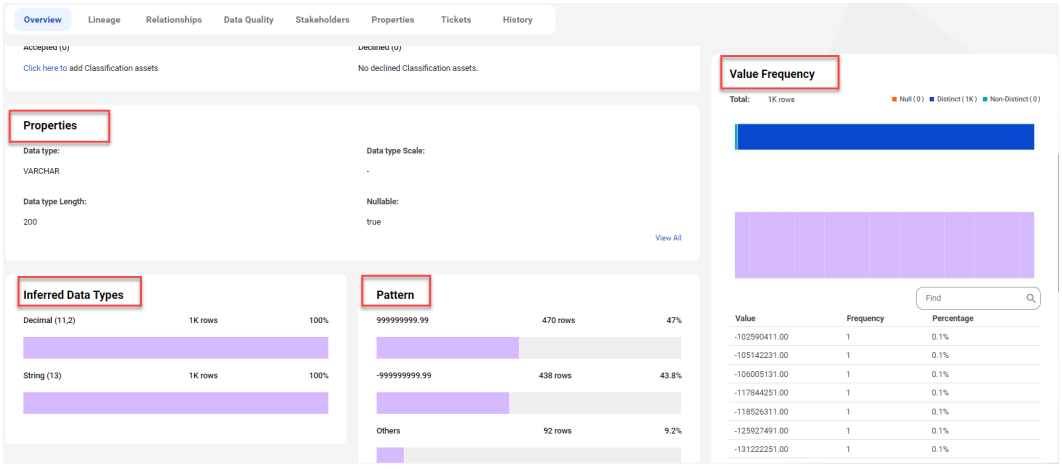
To view data lineage at the data element level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Data Element Level**.

View data profiling results

When you enable the data profiling task for a catalog source in Metadata Command Center, the system runs a profile to evaluate the quality of the metadata extracted from the source system. The profiling statistics appear in Data Governance and Catalog when you open the technical assets.

The scope of profiling statistics that Data Governance and Catalog displays depends on the data profiling configuration parameters that you set when you configured the catalog source in Metadata Command Center.

The following image shows the data profiling statistics that appear on a column asset page in Data Governance and Catalog:



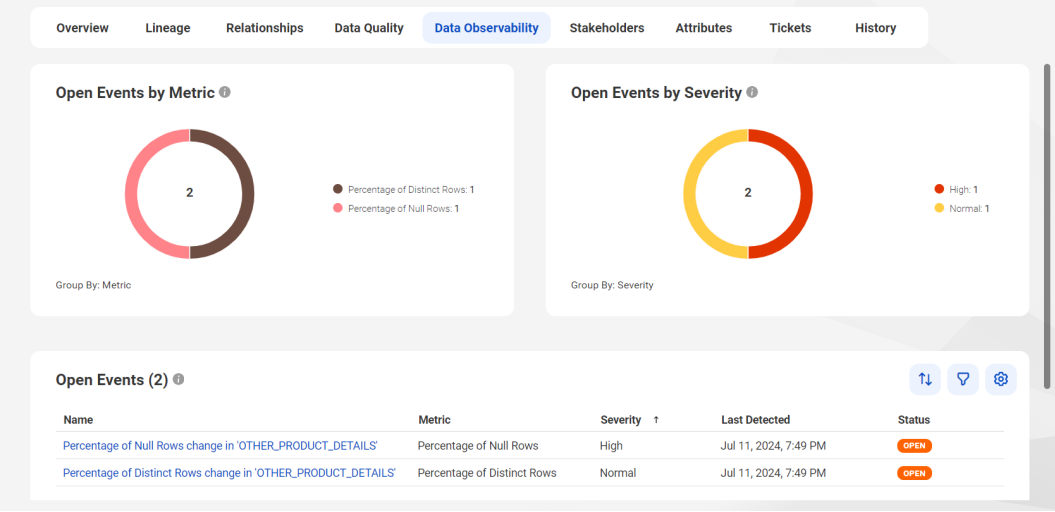
For more information about data profiling results, see *Asset Details* in the Data Governance and Catalog help.

View data observability results

When you enable data observability for a catalog source in Metadata Command Center, you can view and evaluate the events that it generates in Data Governance and Catalog. These events indicate anomalies identified in the characteristics of the profiled data in your source system.

You can view the events that data observability generates for anomalies identified for catalog sources, technical data sets, and data elements. You can then take appropriate actions for the generated events.
Note: The administrator of the catalog source might have applied filters to the data to narrow down the data elements that are applicable for business users in Data Governance and Catalog. The data for which users receive anomaly notifications depend on the filters that are configured for the catalog source.

The following image shows the open events for a column asset in Data Governance and Catalog:

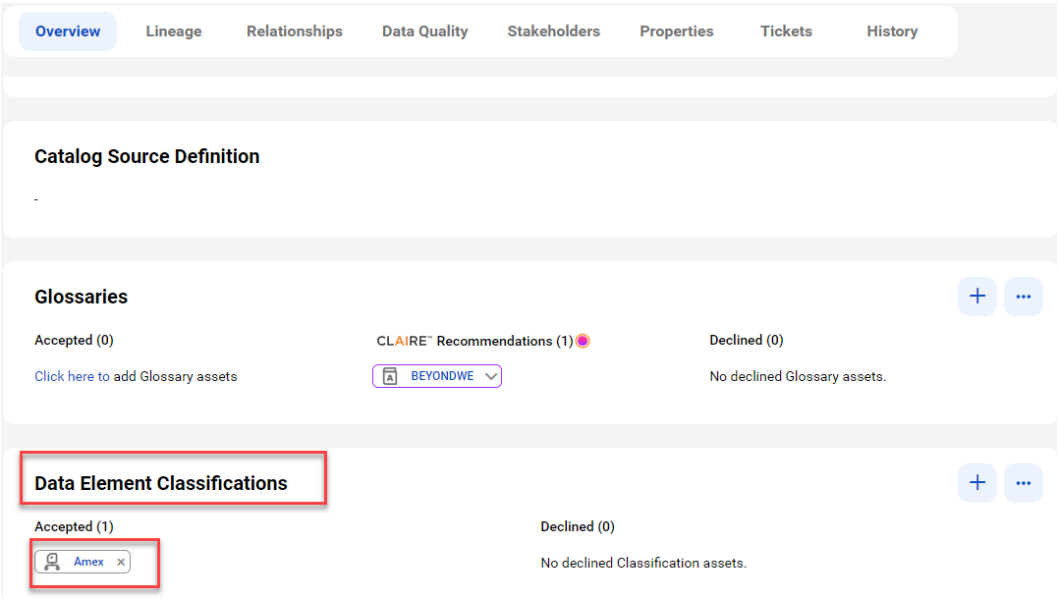


For more information about data observability results, see *Working With Assets* in the Data Governance and Catalog help.

View classified data

When you add data classification rules to a catalog source in Metadata Command Center, the system identifies the columns and tables that match the rules and displays one or more matched data classifications on the column or table asset pages in Data Governance and Catalog.

The following image shows a column asset page with the inferred data element classifications that match the column data and metadata:



For more information about data classification assets, see *Asset Details* in the Data Governance and Catalog help.

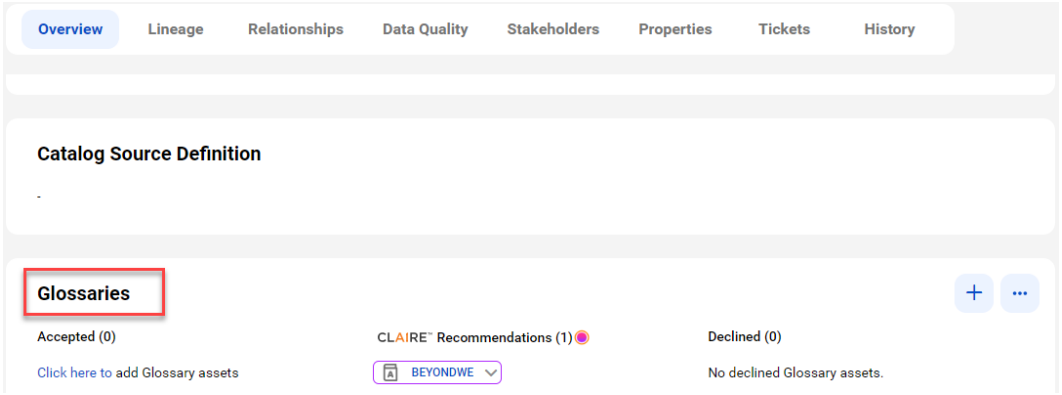
View glossary associations

When you enable the glossary association capability for a catalog source in Metadata Command Center, you can view the accepted glossary assets in Data Governance and Catalog.

The **Overview** tab for a technical asset in the catalog source displays glossary assets in the Accepted and CLAIRE Recommendations sections.

The **Glossaries** panel shows the automatically accepted and CLAIRE® recommended terms.

The following image shows a sample asset page:



APPENDIX A

Managing catalog sources

You can update, copy, delete, purge, and run a catalog source. You can also grant configuration permissions for catalog sources to different roles in your organization.

On the **Catalog Sources** page of the **Explore** page, you can select a catalog source to see its overview and related catalog sources. The **Overview** tab displays the capabilities, filters, stakeholders, asset groups, and schedules enabled for a catalog source. The **Related Catalog Sources** tab displays a list of related catalog sources that are either assigned manually using connection assignment or are curated based on CLAIRE recommendations using the lineage discovery capability. You can view related catalog sources and accept or reject CLAIRE recommendations for catalog sources that are run with the lineage discovery capability.

Cloning a catalog source configuration

You can clone the configuration of a catalog source to create another catalog source with the same configuration.

Note: Cloning doesn't copy the content of the catalog source.

1. In Metadata Command Center, go to the **Explore** page.
2. Select **Catalog Sources** from the menu.
3. You can clone a catalog source in one of the following ways:
   - Hover your mouse over the catalog source that you want to clone and click **Clone** from the **Action** menu.
   - Open the catalog source that you want to clone and click **Clone**.

   Note: By default, the cloned catalog source has the same name suffixed with _Clone. You can change the name if needed.
4. If needed, modify the configuration of the catalog source by updating the parameters on each tab.

   Cloning doesn't copy the content of sensitive fields. If the catalog source contains sensitive fields such as Client Secret, enter the values.
5. Click **Save**.

Deleting or purging a catalog source

Delete or purge a catalog source based on your business requirements.

When you delete a catalog source, you delete all the configuration specific to that catalog source, along with all the metadata that was extracted and any enrichment made to the associated metadata. Purging a catalog source deletes all the metadata that was extracted, along with any enrichment made to the associated metadata.

Note: Before you delete or purge a catalog source, manually unassign the connections that are assigned to that catalog source.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source that you want to delete or purge.
4. Click the **Action** menu, and select **Delete** or **Purge**.

   Note: You can't delete or purge a catalog source if the catalog source job is in the running state.
5. To confirm delete or purge, click **OK**.

   The catalog source is triggered to delete or purge. You can monitor the status of the job on the **Monitor** page.

   Note: If the deletion job succeeds, the catalog source disappears from Metadata Command Center. If it fails, some associated objects might still be present in Data Governance and Catalog. Such catalog sources continue to appear in the list of catalog sources in Metadata Command Center, but are marked as **Deleted**. Delete the catalog sources to remove them along with any remaining objects present in Data Governance and Catalog.

Purging obsolete objects from a catalog source

You can purge obsolete objects from a catalog source.

When you configure or run a catalog source, you can choose to deprecate objects that are deleted from a source system. If you delete objects from the source system or make changes to the filter, the objects imported into the catalog before the change move to the Obsolete lifecycle in the catalog. You can purge obsolete objects from a catalog source. This permanently deletes all objects with the Obsolete lifecycle status, along with associated enrichments, from the catalog.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source from which you want to purge obsolete objects.
4. Click the **Action** menu, and select **Purge Obsolete Objects**.

   A warning message appears on the page.
5. Click **Yes** to confirm.

   A confirmation message appears on the page and the purge job starts. You can monitor the status of the job on the **Monitor** page. If the job succeeds, all obsolete objects present in the catalog source disappear from the catalog.

   Note: If the job fails, some obsolete objects might remain in the catalog. To remove the remaining obsolete objects, run the purge job again.

Curating CLAIRE recommendations

You can view related catalog sources and accept or reject CLAIRE recommendations for a catalog source.

Applicable to catalog sources that generate lineage.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source for which you want to curate CLAIRE recommendations.
4. Hover your mouse over the catalog source and click **View Lineage Discovery** from the **Action** menu.

   Note: The **View Lineage Discovery** option is available for catalog sources that are run with the lineage discovery capability.

   The **Configure** page appears with the catalog source filtered on the **Lineage Discovery** tab of the **Lineage** tab.
   The **Related Catalog Sources** tab displays a list of related catalog sources.
5. Hover your mouse over a related catalog source and click **Accept** or **Reject** from the **Action** menu.

   For more information about lineage discovery and curating CLAIRE recommendations, see *Lineage discovery* in the *Administration* help.

Managing configuration permissions for catalog sources

As a service administrator, you can grant configuration permissions for catalog sources to different roles in your organization, and add specific users and user groups to those roles so that they can perform operational activities on a catalog source.

To manage permissions for catalog source configuration, verify that at least one role has the catalog source configuration **Set Permissions** functionality enabled for Metadata Command Center in Administrator. In Administrator, create the required roles, assign users and user groups to the roles, and specify appropriate permissions to the roles.

You can select roles and add specific users or user groups to those roles to grant the following levels of permission for catalog source configuration:

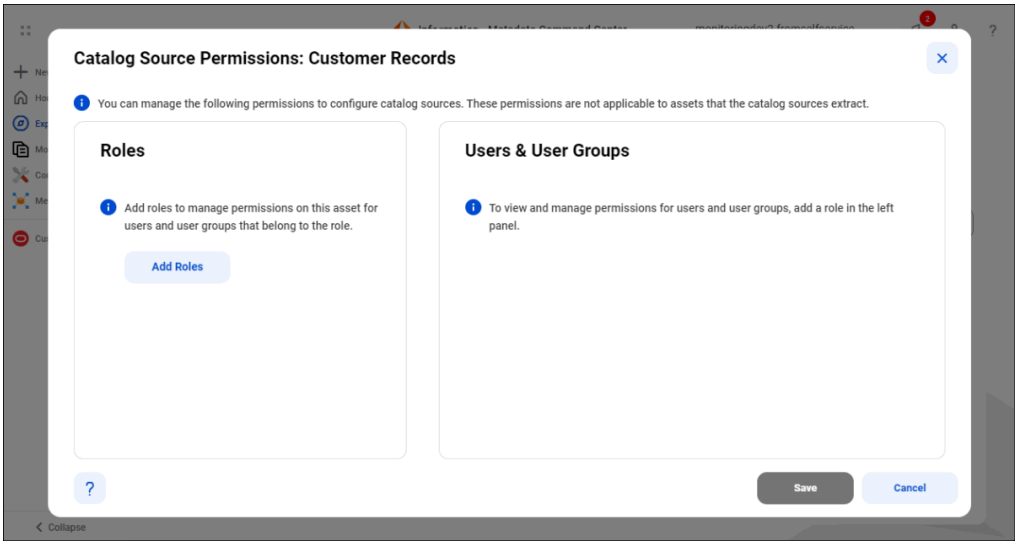
- Create
- Read
- Update
- Run
- Set Permissions

Based on the permissions granted to the roles, the selected users and user groups belonging to that role can perform appropriate operations for configuring catalog sources.

1. Go to the **Explore** page in Metadata Command Center.
2. From the list of catalog sources, search for the catalog source for which you want to manage permissions.

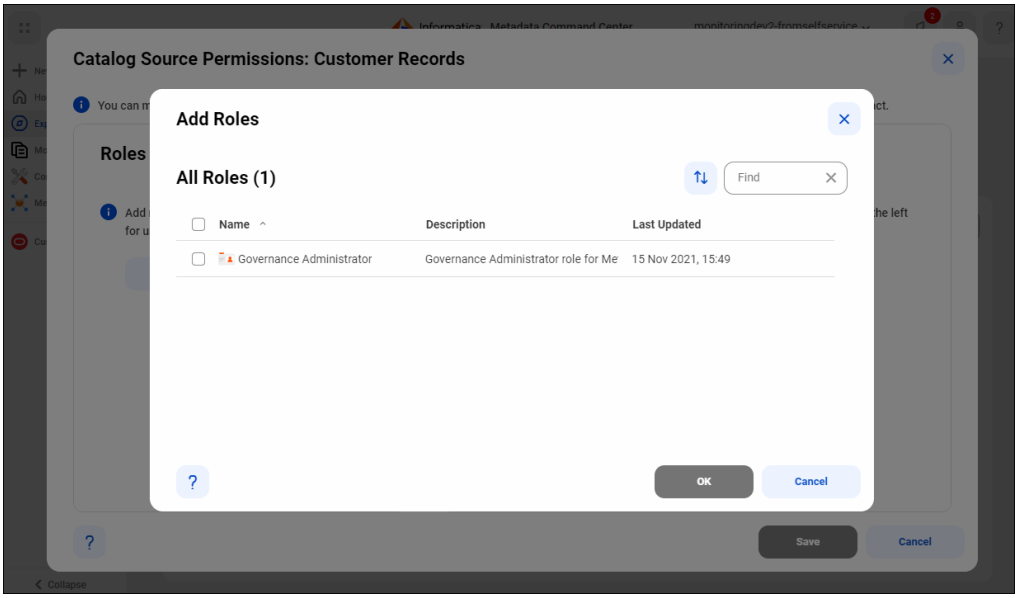
3. Click the **Action** menu, and select **Permissions**.

   The **Catalog Source Permissions** dialog box appears.



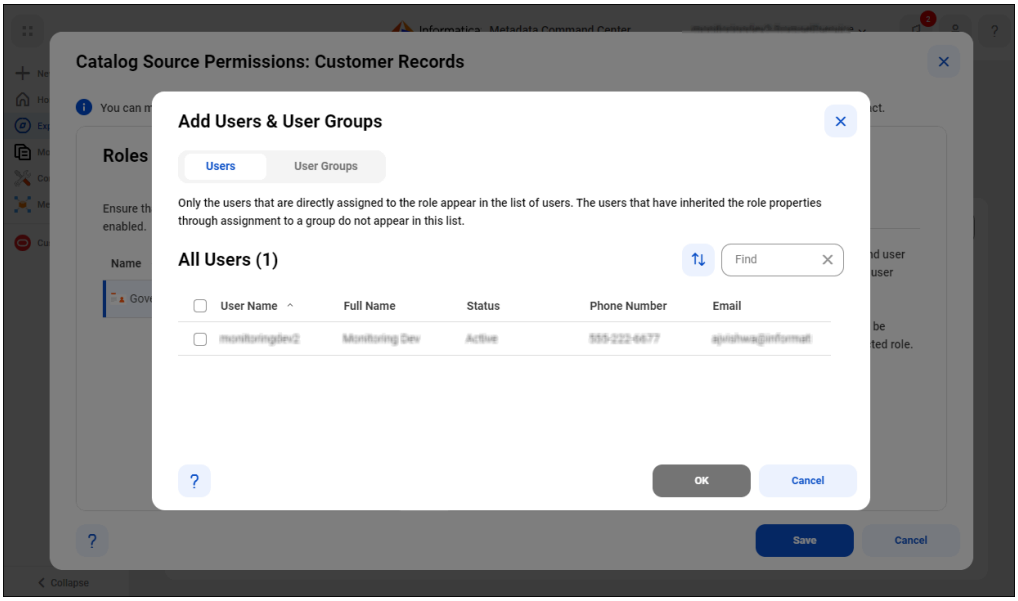
4. In the **Roles** panel, click **Add Roles**.

   The **Add Roles** dialog box appears.



5. Select one or more roles, and click **OK** to add the roles.
6. In the **Add Users & User Groups** panel, by default, all users and user groups that belong to the selected role are granted the role-defined permissions for the catalog source configuration.

To assign permissions only to the selected users and user groups that belong to this role, choose **Specific Users and User Groups**.



7. Select **Add Specific Users and User Groups**.

   The **Add Users & User Groups** dialog box displays the list of users and user groups that the administrator has assigned to the selected role.

8. Select one or more users or user groups to assign to the selected role, and click **OK**.

   Only the selected users and user groups belonging to the specified role will be granted the role-defined permissions for catalog source configuration.

9. View the assigned permissions for each user and or group, and click **Save**.